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EXPERIMENT-5

Question 1

A square park has side **s** meters. Write a C program to find the distance from one corner to the opposite corner.

Step 1: Problem Understanding

Input: Side of the square (s)

Output: Distance between opposite corners (Diagonal)

Formula: Diagonal = Side × 1.414

Step 2: Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1	10	14.14	14.14	Pass
2	15	21.21	21.21	Pass
3	20	28.28	28.28	Pass

Step 3: Algorithm

1. Start the program.
2. Declare the required variables.
3. Read the side of the square.
4. Calculate the diagonal using the formula:**Diagonal = Side** × **√2**
5. Display the diagonal.
6. Stop the program.

Step 4: C Program

```c
#include <stdio.h>

int main()
{
    float s, diagonal;

    printf("Enter the side of square: ");
    scanf("%f", &s);

    diagonal = s * 1.414;

    printf("Distance between opposite corners = %.2f meters", diagonal);

    return 0;
}
```

Step 5: Compilation, Execution

```
┌──(kali㉿kali)-[~]
└─$ gcc hdoc.c -o hdoc

┌──(kali㉿kali)-[~]
└─$ ./hdoc
Enter the side of square: 10
Distance between opposite corners = 14.14 meters

┌──(kali㉿kali)-[~]
└─$ gcc hdoc.c -o hdoc

┌──(kali㉿kali)-[~]
└─$ ./hdoc
Enter the side of square: 15
Distance between opposite corners = 21.21 meters

┌──(kali㉿kali)-[~]
└─$ gcc hdoc.c -o hdoc

┌──(kali㉿kali)-[~]
└─$ ./hdoc
Enter the side of square: 20
Distance between opposite corners = 28.28 meters
```

Question 2

Write a C program to find the remaining area of a parallelogram-shaped field when a rectangular pond is removed from it.

Step 1: Problem Understanding

Input

- Base of parallelogram
- Height of parallelogram
- Length of rectangle

- Breadth of rectangle

Output

- Remaining area of the field

Formula

Area of Parallelogram = Base × Height

Area of Rectangle = Length × Breadth

Remaining Area = Area of Parallelogram − Area of Rectangle

Step 2: Test Case Table

Test Case	Input(Base,Height,Length,Breadth)	Expected Output	Actual Output	Result
1	20,10,5,4	180	180	Pass
2	15,8,3,2	114	114	Pass
3	30,12,6,5	330	330	Pass

Step 3: Algorithm

1. Start the program.
2. Declare all required variables.
3. Read the base and height of the parallelogram.
4. Read the length and breadth of the rectangle.
5. Calculate the area of the parallelogram.
6. Calculate the area of the rectangle.
7. Subtract the rectangle area from the parallelogram area.
8. Display the remaining area.
9. Stop the program.

Step 4: C Program

```c
#include <stdio.h>

int main()
{
    float base, height, length, breadth;
    float parallelogramArea, rectangleArea, remainingArea;

    printf("Enter base and height of parallelogram: ");
    scanf("%f %f", &base, &height);

    printf("Enter length and breadth of rectangle: ");
    scanf("%f %f", &length, &breadth);

    parallelogramArea = base * height;
    rectangleArea = length * breadth;

    remainingArea = parallelogramArea - rectangleArea;

    printf("Remaining Area = %.2f square meters", remainingArea);

    return 0;
}
```

Step 5: Compilation, Execution

```
┌──(kali㉿kali)-[~]
└─$ gcc hdocc.c -o hdocc

┌──(kali㉿kali)-[~]
└─$ ./hdocc
Enter base and height of parallelogram: 20 10
Enter length and breadth of rectangle: 5 4
Remaining Area = 180.00 square meters

┌──(kali㉿kali)-[~]
└─$ gcc hdocc.c -o hdocc

┌──(kali㉿kali)-[~]
└─$ ./hdocc
Enter base and height of parallelogram: 15 8
Enter length and breadth of rectangle: 3 2
Remaining Area = 114.00 square meters

┌──(kali㉿kali)-[~]
└─$ gcc hdocc.c -o hdocc

┌──(kali㉿kali)-[~]
└─$ ./hdocc
Enter base and height of parallelogram: 30 12
Enter length and breadth of rectangle: 6 5
Remaining Area = 330.00 square meters
```

Conclusion

Both C programs were successfully coded, compiled, and executed. The outputs obtained matched the expected results for all test cases. Therefore, both programs are correct and have been verified successfully.